

Sophia Li

Sophomore, Mechanical Engineering | 770-778-1826 | sophiatl@stanford.edu | sophiaylii.github.io

Mechanical Engineering and Aeronautics & Astronautics student at Stanford University with hands-on, full-cycle aerospace hardware experience spanning CAD design, structural/FEA analysis, composite manufacturing, and flight hardware fabrication. Experienced in translating engineering concepts from analysis through manufacturing and testing. Passionate about applying rigorous structural analysis and hands-on fabrication skills to ensure the safety and reliability of commercial aircraft systems.

EDUCATION

Mechanical Engineering, Stanford University

Palo Alto, CA | **09/2025 – 06/2029**

GPA: 4.048

Relevant Coursework: CS223A/ME320 – Introduction to Robotics, AA100 – Introduction to Aeronautics and Astronautics, CME102 – Ordinary Differential Equations for Engineers, CS109 – Probability for Computer Scientists, CS106A/B – Programming (Python, C++)

SKILLS

- **Hardware & Manufacturing:** Flight Hardware Development, Rapid Prototyping, Design for Manufacturing (DFM), Design for Assembly (DFA), Composites Manufacturing (wet layup, foam-core), 3D Printing, Fabrication Tooling, Soldering, Circuit Wiring, Woodworking
- **CAD & FEA:** Autodesk Fusion 360, Abaqus (FEA), GD&T, HyperX (Collier Aerospace)
- **Structural Analysis:** Classical Laminate Theory, Beam Theory, Composite Structures, Mechanics of Materials, Failure Criteria
- **Aerodynamics:** XFLR5, Airfoil Analysis, Lift Distribution Modeling, Wing Sizing
- **Programming:** Python, C++, MATLAB; proficient in Microsoft Office (Excel, Word, PowerPoint)

ENGINEERING PROJECTS

Composite Wing Design, Analysis, Manufacturing, & Testing

06/2026 – 07/2026

- Designed and manufactured a carbon fiber composite foam-core NACA 2412 wing in Fusion 360, carrying the full structural design cycle from aerodynamic loading through physical prototype and test.
- Fabricated flight-representative composite wing hardware using wet layup and foam-core manufacturing techniques and built 2D/3D Abaqus composite shell models validating structural response within 2% of analytical predictions.
- Performed structural failure analysis using HyperX Tsai-Wu criteria to identify laminate failure modes and evaluate design margins, the same margin-of-safety approach used in assessing composite structural repairs and damage tolerance.
- Developed MATLAB-based tools to evaluate laminate stiffness, neutral axis location, and bending response, informing layup and spar design decisions.

Custom Hot Wire Cutter Design & Fabrication

07/2026

- Independently designed and fabricated a nichrome-wire hot wire cutter integrating electrical power delivery, thermal control, and mechanical structure for precision foam-core manufacturing.
- Engineered a tensioning mechanism to maintain straight cutting geometry across long-span foam sections.
- Applied electrical systems, heat transfer, mechanical design, and fabrication knowledge to build a custom manufacturing tool that enabled rapid prototyping of aerodynamic foam cores for composite aircraft structures.

EXPERIENCE

Mechanical Design Lead, Stanford Design/Build/Fly

Stanford University | **09/2025 – Present**

- Leads mechanical design and prototyping for an AIAA competition aircraft, managing 10 engineers across design, manufacturing, and testing.
- Designed, fabricated, and tested a two-stage servo-actuated mechanical release mechanism, a compact electromechanical assembly integrating servos with flight hardware requirements.
- Translated aircraft performance requirements into manufacturable mechanical subsystem designs, managing full aircraft CAD integration in Fusion 360 across fuselage, wing, and control surfaces.
- Created manufacturing documentation and supported design reviews to accelerate prototype development and fabrication cycles.

Research Assistant, Stanford Freeman Spogli Institute (FSI)

Stanford University | **04/2026 – Present**

- Sole author of the lab's IRB application; facilitated technical communication between faculty leads and executed quantitative analysis of large-scale survey data, developing visualizations to communicate findings to collaborators.

VOLUNTEER EXPERIENCE

Outreach, Stanford STEM Nights

Stanford University | **10/2025 – Present**

- Coordinated cross-functional teams to deliver technical demonstrations for large-scale STEM engagement events, ensuring safety compliance and effective public communication.

AWARDS

- 2024 + 2025 Morley Award Recipient (~23/1161) — awarded to Georgia Tech's distance math students whose final grades in linear algebra and multivariable calculus ranked in the top 2% of the class.